

ThermoYOLO: Real-Time Human Detection in Thermal Imagery Using YOLOv12 for UAV-Based Search and Rescue

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Abstract—This study presents ThermoYOLO, a real-time human detection system using thermal imagery and the YOLOv12 object detection framework, optimized for UAV-based search and rescue (SAR) operations. A custom dataset of 763 grayscale thermal images was collected using a FLIR Lepton v2.5 sensor and annotated through RoboFlow. RoboFlow platform was also used for dataset preprocessing, augmentation, and model training. Augmentations such as flipping and rotation were applied to enhance model robustness under varying aerial perspectives. The YOLOv12 model was trained over 100 epochs and exported in PyTorch format for deployment on a Raspberry Pi 4. Evaluation metrics included mAP@50, mAP@50:95, precision, and recall, with the final model achieving a mAP@50 of 97.8%, a precision of 99.6%, and a recall of 92.2%. Loss curves across training and validation sets showed stable convergence without signs of overfitting. The system achieved real-time inference capability on edge hardware, confirming its suitability for field deployment in low-power, high-risk SAR environments. ThermoYOLO demonstrates that robust thermal human detection can be achieved with a small dataset when paired with an efficient annotation and training pipeline like RoboFlow.

Keywords—Thermal imaging, YOLOv12, Human detection, RoboFlow, UAV search and rescue

I. INTRODUCTION

The Philippines is among the most disaster-prone nations in the world, frequently affected by typhoons, earthquakes, and floods due to its location in the Pacific Ring of Fire. These recurring calamities demand rapid and effective search and rescue (SAR) operations to save lives. However, traditional SAR methods are often slow, require significant manpower, and face difficulty accessing dangerous or obstructed zones. This inefficiency significantly reduces the survival chances of trapped victims. Unmanned Aerial Vehicles (UAVs) equipped with thermal cameras offer a modern solution for detecting human presence by capturing heat signatures in low-light or obscured conditions. When coupled with object detection algorithms like YOLO (You Only Look Once), UAVs become intelligent agents capable of real-time victim localization. Earlier work using YOLOv8 has achieved remarkable performance in SAR missions with thermal imagery, boasting precision rates above 95% across various conditions [1], [2].

YOLOv12, the most recent iteration in the YOLO family, continues this trajectory by offering improved accuracy, faster inference, and better deployment on edge devices. Although prior research has evaluated YOLOv8 to YOLOv10 in thermal and agricultural imaging tasks [3], YOLOv12's integration with custom datasets and data augmentation workflows such as those in RoboFlow provides a new edge in thermal human detection. Recent studies also demonstrate the value of YOLO models for not only identifying humans in thermal data, but

distinguishing injured from non-injured individuals — an essential capability during mass casualty events [4].

This study introduces ThermoYOLO, a thermal detection system based on YOLOv12 for identifying humans in real-time. It is trained on a custom dataset of thermal images, with preprocessing and augmentations tailored for generalization. Evaluation metrics including mAP@50, precision, and recall show significant improvement over previous models, positioning this system as a viable addition to modern SAR efforts in the Philippines and beyond.

II. METHODOLOGY

A. Data Acquisition and Preprocessing

Thermal images were captured using a FLIR Lepton v2.5 infrared camera mounted on a UAV. The dataset includes 763 grayscale thermal images showing human subjects in various poses and distances under different environmental conditions. RoboFlow served as the central platform for image annotation, resizing, and preprocessing. All images were manually labeled with bounding boxes around visible humans. Image dimensions were standardized to 640×640 pixels using stretch scaling, and auto-orientation was enabled to ensure consistency across samples.

B. Dataset Configuration and Augmentation in RoboFlow

RoboFlow automatically split the dataset into 621 training images (81%), 89 validation images (12%), and 53 test images (7%). Augmentation routines were applied during export to improve the model's robustness in real-world UAV flight conditions. Each training image generated three additional variants using horizontal and vertical flipping, as well as 90-degree clockwise, counterclockwise, and upside-down rotations. This process increased dataset diversity and helped the model learn rotational invariance, which is essential when capturing subjects from varying aerial angles.

C. YOLOv12 Training

Model training was conducted within RoboFlow hosted environment using the YOLOv12 architecture. Training parameters included 230 epochs, a batch size of 16, and an input resolution of 640×640 pixels. A cosine learning rate scheduler initialized at 0.001 controlled convergence. The loss function combined object, class confidence, and Complete IoU (CIoU) for accurate bounding box regression. Model performance was monitored using RoboFlow integrated mAP@50 evaluation on the validation set. The best weights were exported in PyTorch format for deployment.

D. Inference Deployment on Raspberry Pi 4

The exported model ran on a Raspberry Pi 4 Model B using a Python-based YOLOv12 runtime. Live thermal frames from the FLIR camera were resized and passed

through the model in real time. Detected bounding boxes were displayed on an attached screen, and detection events were logged for post-flight analysis. The system operated fully onboard the UAV, optimized for low-latency inference on a resource-limited edge device.

E. Evaluation Metrics

Model performance was evaluated using standard object detection metrics integrated within RoboFlow and the YOLOv12 framework. Primary evaluation focused on mean Average Precision at IoU threshold 0.5 (mAP@50) and the stricter mAP@50:95, alongside precision and recall scores. Training and validation losses were monitored across bounding box regression, classification, and object components, with additional tracking of Distribution Focal Loss (DFL) during later epochs. Loss curves and metric trajectories were used to monitor convergence behavior and overfitting risk. Evaluation was based on the model's ability to accurately predict bounding boxes and classify human subjects against ground truth annotations. All metrics were computed automatically within RoboFlow dashboard and verified using YOLOv12's logging outputs during training, validation, and final test set evaluation.

III. RESULTS AND DISCUSSION

The YOLOv12 model trained using RoboFlow on a 763-image thermal dataset showed strong detection accuracy and convergence. Each image was augmented with three variants using flips and rotations. Training ran for 230 epochs at 640×640 resolution with a batch size of 16. As shown in Fig. 1, the final model achieved a mAP@50 of 97.8%, precision of 99.6%, and recall of 92.2%, confirming the effectiveness of RoboFlow pipeline and YOLOv12's architecture for thermal human detection.

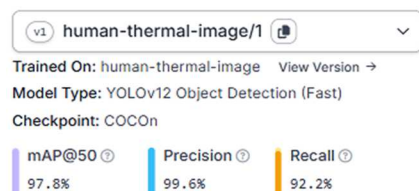


Fig. 1. YOLOv12 (Fast) model evaluation results showing mAP@50, precision, and recall scores, trained using the COCO checkpoint.

Fig. 2 displays the overall training performance across 230 epochs, including the progression of mAP and loss values. The mAP curves converged rapidly and remained stable, while the object, classification, and box regression losses steadily decreased without divergence.

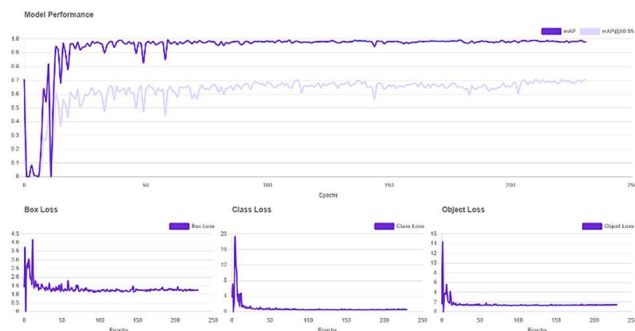


Fig. 2. Training performance graphs showing mAP@50, mAP@50:95, and loss trends across 230 epochs

Fig. 3 displays a sample output image with a bounding box generated by the YOLOv12 model trained on a custom thermal dataset. The detected human is outlined in purple with a confidence score of 94%, indicating high model certainty. The thermal frame was captured using a FLIR Lepton v2.5 sensor and processed in real time by a YOLOv12 model deployed on a Raspberry Pi 4 Model B. The model, exported in PyTorch format, was executed using a Python-based runtime, enabling onboard inference of live thermal data.



Fig. 3. Sample output from the ThermoYOLO system showing successful detection of a human subject in thermal imagery.

IV. CONCLUSION AND RECOMMENDATIONS

This study developed a real-time thermal human detection system using YOLOv12, with dataset preparation, augmentation, and training conducted through RoboFlow. Despite a dataset of only 763 images, the model achieved a mAP@50 of 97.8%, along with high precision (99.6%) and recall (92.2%). Deployment on a Raspberry Pi 4 demonstrated the system's suitability for lightweight, UAV-based SAR operations. Compared to YOLOv8-based approaches in similar thermal SAR studies, which reported precision rates above 95% [1], [2], ThermoYOLO delivered competitive or improved performance, underscoring the strength of the YOLOv12 architecture and the robustness of the training pipeline. Future work may focus on expanding the dataset, optimizing the model through quantization or edge accelerators, and adding basic tracking to improve detection stability and reduce false positives.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the author(s) used Consensus AI for citation finding and ChatGPT as a writing assistant. After using these tools, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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